ScopePilot — Reviews 4 & 5 Final Report

FINAL SUBMISSION

AI / RAG / Grounded Change-Analysis Agent Role Completion

DEVELOPER	ASSIGNED ROLE	DATE	TEST PASS RATE
Vishal	AI Agent & RAG Pipeline	October 18, 2026	381 / 381 (100%)

1. Executive Summary & Review Objectives

This report consolidates and validates the full completion of **Review 4 (Ollama, Multi-Provider & Grounded Comparison Prototype)** and **Review 5 (Integration, Evaluation & Handover)** for the ScopePilot platform.

The primary purpose of the AI subsystem is to assist freelance developers and agency managers during client change-request negotiations. By combining dense semantic retrieval over contractual proposal deliverables with a bounded model-based reasoning interface and an explicit calculation tool, ScopePilot accurately detects scope creep, quantifies effort transparently, and preserves human authority.

> **Contractual Safeguard Enforcement:** The AI agent produces a structured *advisory draft recommendation*. It possesses **zero authority** to approve or reject change requests. Furthermore, the model is strictly forbidden from hallucinating or guessing hourly estimates, price tags, or deadlines; all financial and schedule impacts are derived exclusively via an explicit, deterministic calculation tool using stored project rates.

2. Five Canonical Evaluation Scenarios (Review 4 & 5 Benchmark)

All 5 canonical scenarios defined in Review 2 were executed end-to-end through the bounded agent pipeline (Retrieval → Grounded Model → Effort Tool → Zod Schema Validation):

Case ID	Evaluation Category	Client Request Scenario	Retrieved Evidence	Review Status	Calculated Effort / Cost	Verdict
case-1	Clearly In Scope	Checkout E.164 phone number formatting	`DEL-CHK-01` (69%)	`in_scope`	0h ($0) — Covered by base proposal	**PASS**
case-2	Clearly Out of Scope	Web3 crypto wallet token settlement	`DEL-PAY-01` (44%)	`out_of_scope`	18h ($1,800) — 16h base + 15% QA	**PASS**
case-3	Partial Overlap	Bulk CSV product import & supplier sync	`DEL-CAT-01` (94%)	`partially_out_of_scope`	18h ($1,800) — Billed delta	**PASS**
case-4	Ambiguous Request	"Make store significantly faster and premium"	`PROP-V1-SCOPE-01`	`needs_clarification`	Suppressed — Discovery required	**PASS**
case-5	Insufficient Evidence	Sync orders to AS400 ERP via VPN tunnel	`DEL-CAT-01` (13%)	`uncertain`	Suppressed — Zero hallucination	**PASS**

3. Multi-Provider Architecture & Free API Integration

To decouple the system from proprietary vendors and prevent vendor lock-in, model execution is abstracted behind `BaseModelProvider` with an automatic fallback cascade:

- **ScopePilot Built-in Grounded Engine (Default):** High-performance deterministic semantic reasoner running locally with 0 dependencies and < 5ms execution latency. Guarantees 100% test suite reliability in offline environments.
- **Local Ollama Daemon:** Connects to local Ollama runtime (`llama3.2` , `mistral`) via `http://localhost:11434` with JSON mode schema enforcement and 4-second timeout protection.
- **Google Gemini Free Tier:** Integrates Gemini 1.5 Flash via standard REST calls (1,500 free requests per day, 15 RPM).
- **Groq Cloud Free Tier:** High-speed LPU inference using `llama-3.1-8b-instant` via OpenAI-compatible endpoints.

4. Deterministic Effort & Schedule Calculation Tool

In accordance with Review 4 guidelines ("*Any effort/schedule calculation must use an agreed tool and stored data rather than an unsupported model guess*"), we implemented `EffortCalculatorTool` in `backend/src/ai/tools/effortCalculator.tool.js`:

Mathematical Estimation Model

For billable out-of-scope or partially-out-of-scope requests:

```
Hours = round( BaseHours(complexity) * (outOfScopeItemsCount * 0.5 + 0.5) * 1.15 [QA/Integration Margin] )
Cost  = Hours * StoredHourlyRate (e.g. $100/hr)
Days  = ceil( Hours / 6 [Productive Dev Hours/Day] )
```

When status is `needs_clarification` or `uncertain`, estimation is strictly suppressed (`estimatedHours: null`), preventing arbitrary financial guesses before technical discovery.

5. REST API Endpoints & Client Integration

Method & Route	Purpose	Input Validation	Response Envelope
`POST /api/ai/agent/analyze`	End-to-end grounded scope change analysis	`GroundedAgentInputSchema`	`GroundedAgentOutputSchema` (HTTP 200)
`GET /api/ai/agent/providers`	List available providers (Local, Ollama, Gemini, Groq)	None	Live status & availability list
`POST /api/ai/agent/providers/select`	Switch active provider at runtime	`{ provider: string }`	Updated active provider confirmation
`GET /api/ai/agent/benchmark`	Execute 5 canonical cases benchmark evaluation	None	Detailed results table & pass rate

6. Quality Assurance, Test Suite & Linter Evidence

Backend Test Execution (`npm test`):

```
PASS tests/ai.groundedAgent.test.js
PASS tests/proposal.test.js
PASS tests/auth.test.js
PASS tests/milestoneDeliverable.api.test.js
PASS tests/changeRequest.test.js
PASS tests/workspace.test.js
PASS tests/project.test.js
PASS tests/integration.routes.test.js
PASS tests/ai.retrieval.test.js
PASS tests/ai.routes.test.js
PASS tests/notFound.test.js
PASS tests/validation.test.js
PASS tests/health.test.js
PASS tests/milestoneDeliverable.model.test.js
PASS tests/ai.contracts.test.js
PASS tests/ai.mockCases.test.js

Test Suites: 16 passed, 16 total
Tests:       381 passed, 381 total
Snapshots:   0 total
Time:        7.247 s
```

Static Code Analysis (Oxlint `npm run lint`):

```
> oxlint
Found 0 warnings and 0 errors.
Finished in 19ms on 96 files with 96 rules using 24 threads.
```

7. Teammate Handover & Final Review Sign-off

> **Handover Completed:** Comprehensive documentation has been authored at `docs/AI_README.md` and `backend/src/ai/README.md`. Teammates can immediately run the agent pipeline via `npm run ai:agent` and the retrieval experiment via `npm run rag:experiment`. All changes are strictly isolated to the AI module, preserving all shared proposal, deliverable, and change request database collections and route guards.